

Lecture 3 Homework Solution

Problem 1. Copper loss & Torque constant

$$P_{in} = v_{as} i_a + v_{bs} i_b + v_{cs} i_c = \mathbf{v}_{abcs}^T \mathbf{i}_{abcs} = \mathbf{v}_{dqs}^{rT} \mathbf{T}_{inv}(\theta_r)^T \mathbf{T}_{inv}(\theta_r) \mathbf{i}_{dqs}^r = \frac{3}{2} \mathbf{v}_{dqs}^{rT} \mathbf{i}_{dqs}^r = \frac{3}{2} (v_{ds}^r i_{ds}^r + v_{qs}^r i_{qs}^r)$$

$$\begin{cases} v_{ds}^r = R_s i_{ds}^r + \frac{d\lambda_{ds}^r}{dt} - \omega_r \lambda_{qs}^r \\ v_{qs}^r = R_s i_{qs}^r + \frac{d\lambda_{qs}^r}{dt} + \omega_r \lambda_{ds}^r \end{cases} \quad \textcircled{1}$$

steady state 일때 0.

$$\begin{cases} \lambda_{ds}^r \approx L_s i_{ds}^r + \phi_f \\ \lambda_{qs}^r \approx L_s i_{qs}^r \end{cases} \quad \textcircled{2}$$

core saturation에 의해,
식 ②는 근사식이다.

$$P_{in} = \frac{3}{2} (v_{ds}^r i_{ds}^r + v_{qs}^r i_{qs}^r) = \frac{3}{2} \{ (R_s i_{ds}^r - \omega_r \lambda_{qs}^r) i_{ds}^r + (R_s i_{qs}^r + \omega_r \lambda_{ds}^r) i_{qs}^r \}$$

$$= \frac{3}{2} R_s (i_{ds}^{r2} + i_{qs}^{r2}) + \omega_r \frac{3}{2} (\lambda_{ds}^r i_{qs}^r - \lambda_{qs}^r i_{ds}^r)$$

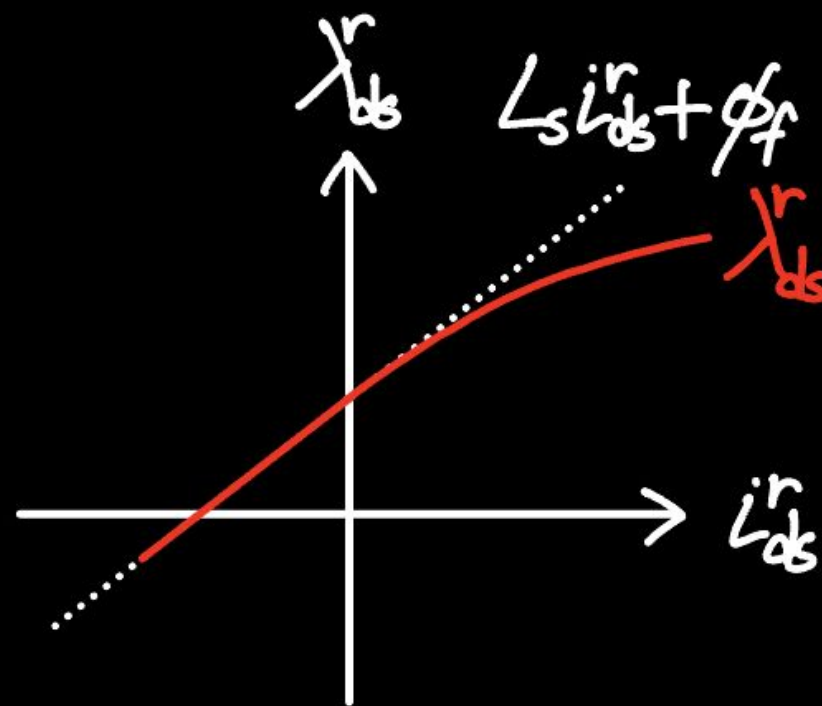
$$\omega_r = \frac{P}{2} \omega_m \text{ 이므로, } \tau = \frac{3}{2} \frac{P}{2} (\lambda_{ds}^r i_{qs}^r - \cancel{\lambda_{qs}^r i_{ds}^r}) = \frac{3}{2} \frac{P}{2} \lambda_{ds}^r i_{qs}^r \quad \textcircled{3}$$

보통 $i_{ds}^r = 0$ 으로 제어.

Problem 1. Copper loss & Torque constant

$$\tau = \frac{3}{2} \frac{P}{2} (\lambda_{ds}^r i_{ds}^r - \cancel{\lambda_{ds}^r i_{ds}^r}) = \frac{3}{2} \frac{P}{2} \boxed{\lambda_{ds}^r} i_{ds}^r$$

보통 $i_{ds}^r = 0$ 으로 제어.

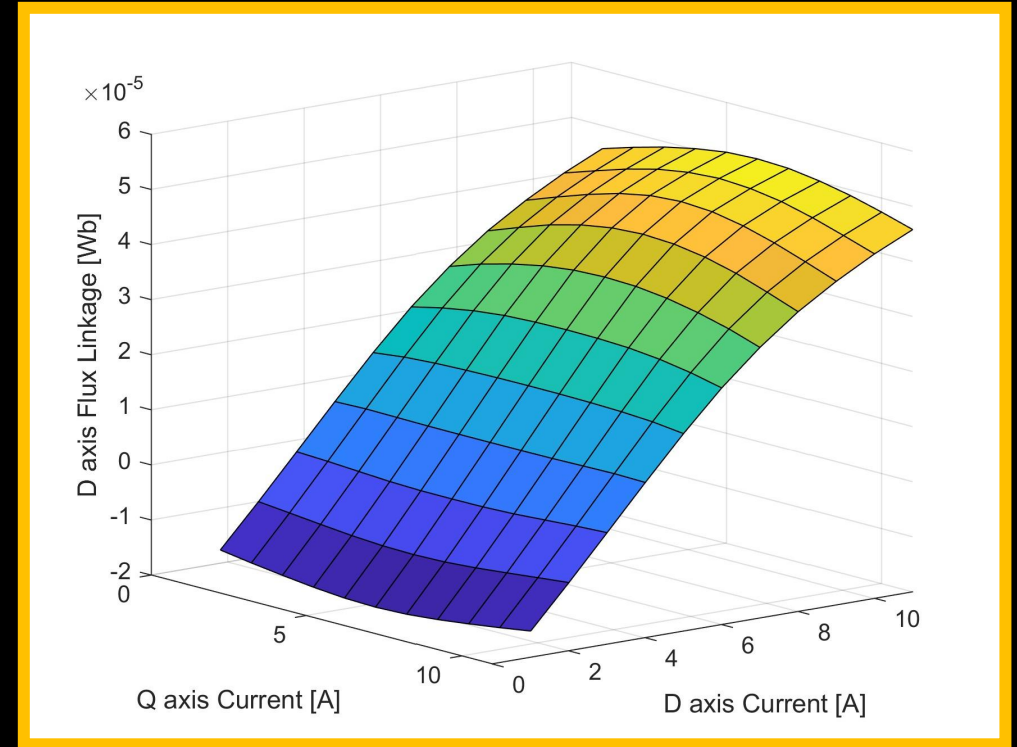


$$\lambda_{ds}^r \approx L_s i_{ds}^r + \phi_f$$

Problem 1. Copper loss & Torque constant

$$\tau = \frac{3}{2} \frac{P}{2} (\lambda_{ds}^r i_{qs}^r - \cancel{\lambda_{qs}^r i_{ds}^r}) = \frac{3}{2} \frac{P}{2} \lambda_{ds}^r i_{qs}^r$$

③
보통 $i_{ds}^r = 0$ 으로 제어.



$$\lambda_{ds}^r \stackrel{\textcircled{2}}{\approx} L_s i_{ds}^r + \phi_f$$

Problem 2. Motor Operating Region

- 이전 과제에서 유도한 SPMSM voltage equation을 이용하여, Motor Operating Region을 matlab으로 plot하라.
 - + 발열량에 의한 제한은 고려하지 않는다.
 - + 자속 포화에 의한 비선형성은 무시한다.

- [SPEC]

$$V_{DC} = 100.8V$$

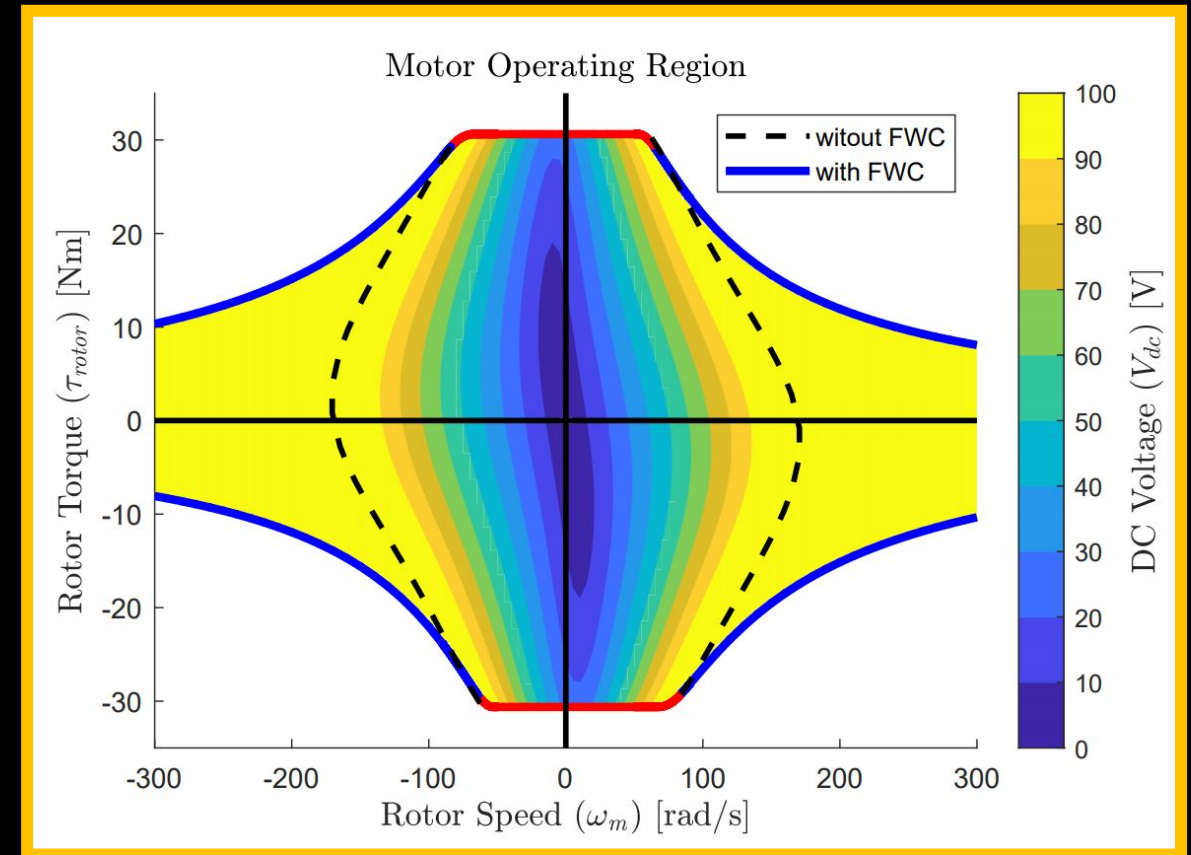
$$\text{Number of Poles} = 30$$

$$R_{tt} = 0.42\Omega$$

$$L_{tt} = 1.58mH$$

$$K_t = 0.5881Nm/A$$

(tt: Terminal-to-Terminal. 3상선 임의의 두 단자 간)



- D축 전류가 0일 때 MOR를 구하라.
- D축 전류는 0이 최선인가? 아니라면 각속도별로 최적의 D축 전류를 구하고 그 때의 MOR을 구하라.

Problem 2. Motor Operating Region

$$\begin{cases} V_{ds}^r = R_s i_{ds}^r + L_s \frac{di_{ds}^r}{dt} - \omega_e L_s i_{qs}^r \\ V_{qs}^r = R_s i_{qs}^r + L_s \frac{di_{qs}^r}{dt} + \omega_e L_s i_{ds}^r + \omega_e \phi_f \end{cases}$$

steady state 일때 0.

전압원의 전압이 V_{dc} 일때, $\sqrt{V_{ds}^2 + V_{qs}^2} \leq \frac{V_{dc}}{\sqrt{3}}$. boundary는 $\sqrt{V_{ds}^2 + V_{qs}^2} = \frac{V_{dc}}{\sqrt{3}}$

$$\begin{aligned} V_{ds}^2 + V_{qs}^2 - \frac{V_{dc}^2}{3} &= (R_s i_{ds} - \omega_e L_s i_{qs})^2 + (\omega_e L_s i_{ds} + R_s i_{qs} + \omega_e \phi_f)^2 - \frac{V_{dc}^2}{3} \\ &= (R_s^2 + \omega_e^2 L_s^2)(i_{ds}^2 + i_{qs}^2) + 2R_s \omega_e \phi_f i_{qs} + \omega_e^2 \phi_f^2 - \frac{V_{dc}^2}{3} = 0 \\ &\quad + 2L_s \omega_e^2 \phi_f i_{ds} \end{aligned}$$

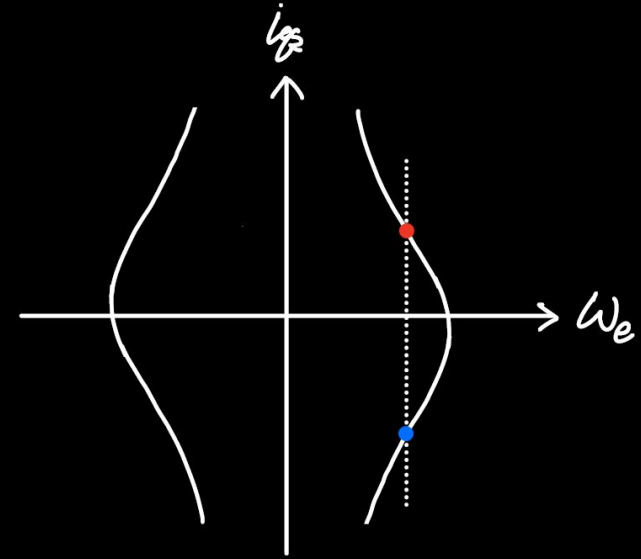
Problem 2. Motor Operating Region

- D축 전류가 0일 때 MOR를 구하라.

$i_{ds} = 0$ 인 경우,

$$(R_s^2 + \omega_e^2 L_s^2) i_{qs}^2 + 2R_s \omega_e \phi_f i_{qs} + \omega_e^2 \phi_f^2 - \frac{V_{de}^2}{3} = 0$$

$$= A(\omega_e) i_{qs}^2 + B(\omega_e) i_{qs} + C(\omega_e) = 0. \quad i_{qs}(\omega_e) = \frac{-B(\omega_e) \pm \sqrt{B^2(\omega_e) - 4A(\omega_e)C(\omega_e)}}{2A(\omega_e)}$$



Problem 2. Motor Operating Region

- D축 전류는 0이 최선인가? 아니면 각속도별로 최적의 D축 전류를 구하고 그 때의 MOR을 구하라.

$i_{ds} \neq 0$ 인 경우,

i_{ds}, i_{qs} 에 대한 원의 방정식이다.

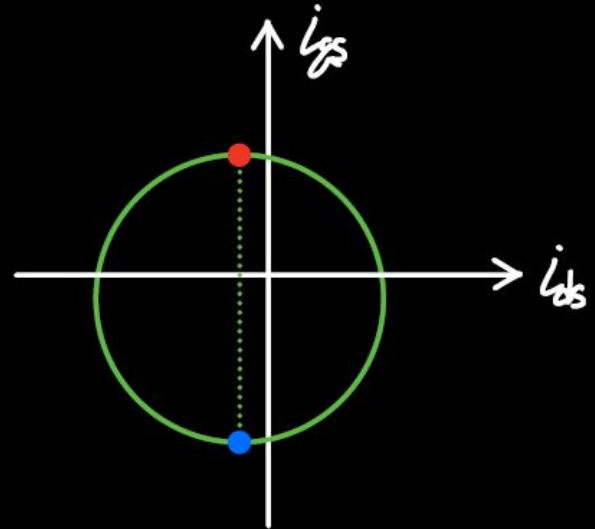
$$2i_{ds}^2 + 2i_{qs}^2 - \frac{V_{dc}^2}{3} = (R_s^2 + \omega_e^2 L_s^2)(i_{ds}^2 + i_{qs}^2) + 2R_s \omega_e \phi_f i_{qs} + \omega_e^2 \phi_f^2 - \frac{V_{dc}^2}{3} = 0.$$

$$+ 2L_s \omega_e^2 \phi_f i_{ds}$$

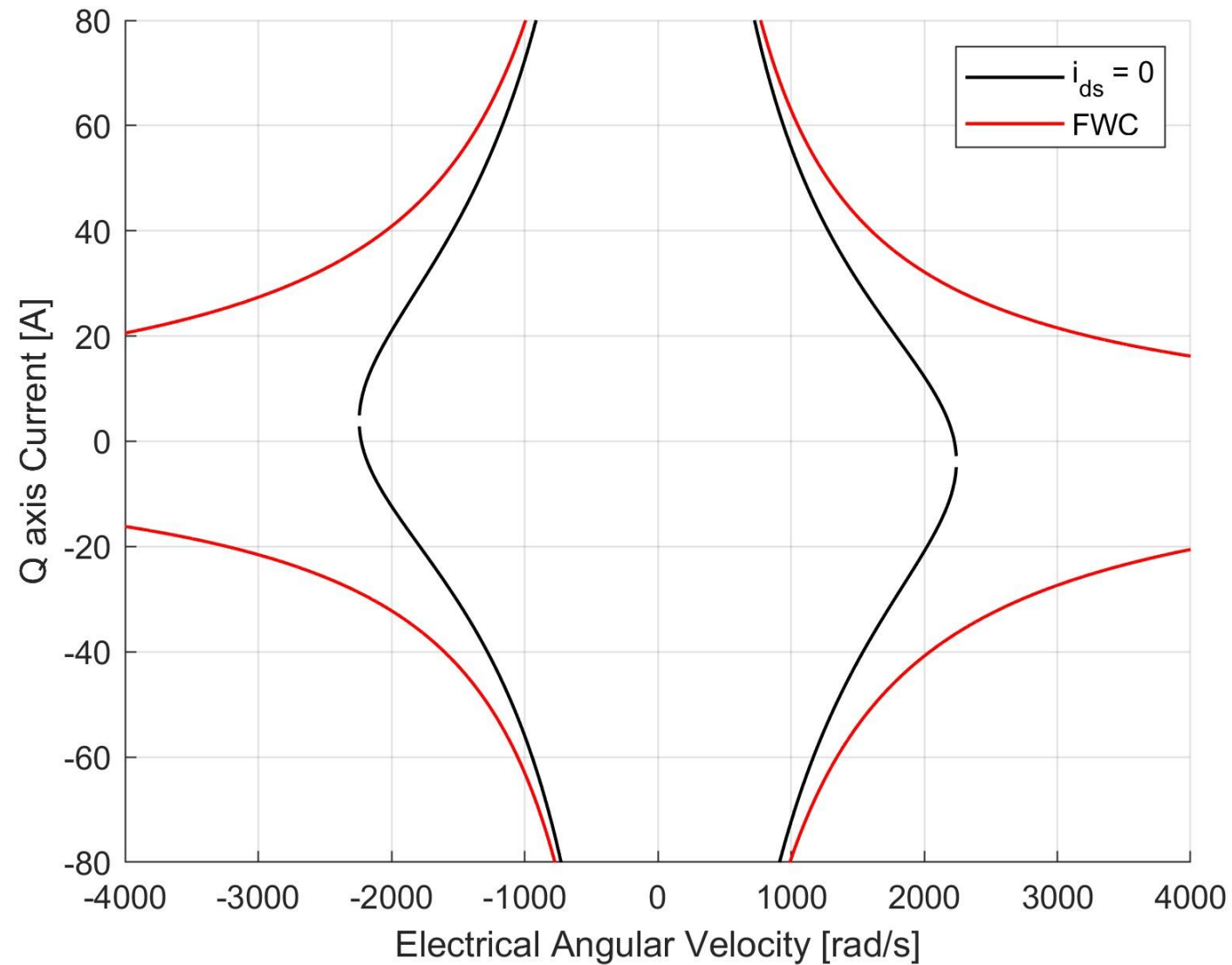
$$= A(\omega_e)(i_{ds}^2 + i_{qs}^2) + B(\omega_e)i_{ds} + C(\omega_e)i_{qs} + D(\omega_e) = 0$$

$$\rightarrow (i_{ds} + \frac{B}{2A})^2 + (i_{qs} + \frac{C}{2A})^2 = \frac{B^2 + C^2}{4A^2} - \frac{D}{A}$$

$$i_{ds}(\omega_e) = -\frac{B(\omega_e)}{2A(\omega_e)} \quad i_{qs}(\omega_e) = -\frac{C(\omega_e)}{2A(\omega_e)} \pm \sqrt{\frac{B(\omega_e)^2 + C(\omega_e)^2}{4A(\omega_e)^2} - \frac{D(\omega_e)}{A(\omega_e)}}$$



Problem 2. Motor Operating Region



Problem 3. Resistive Braking

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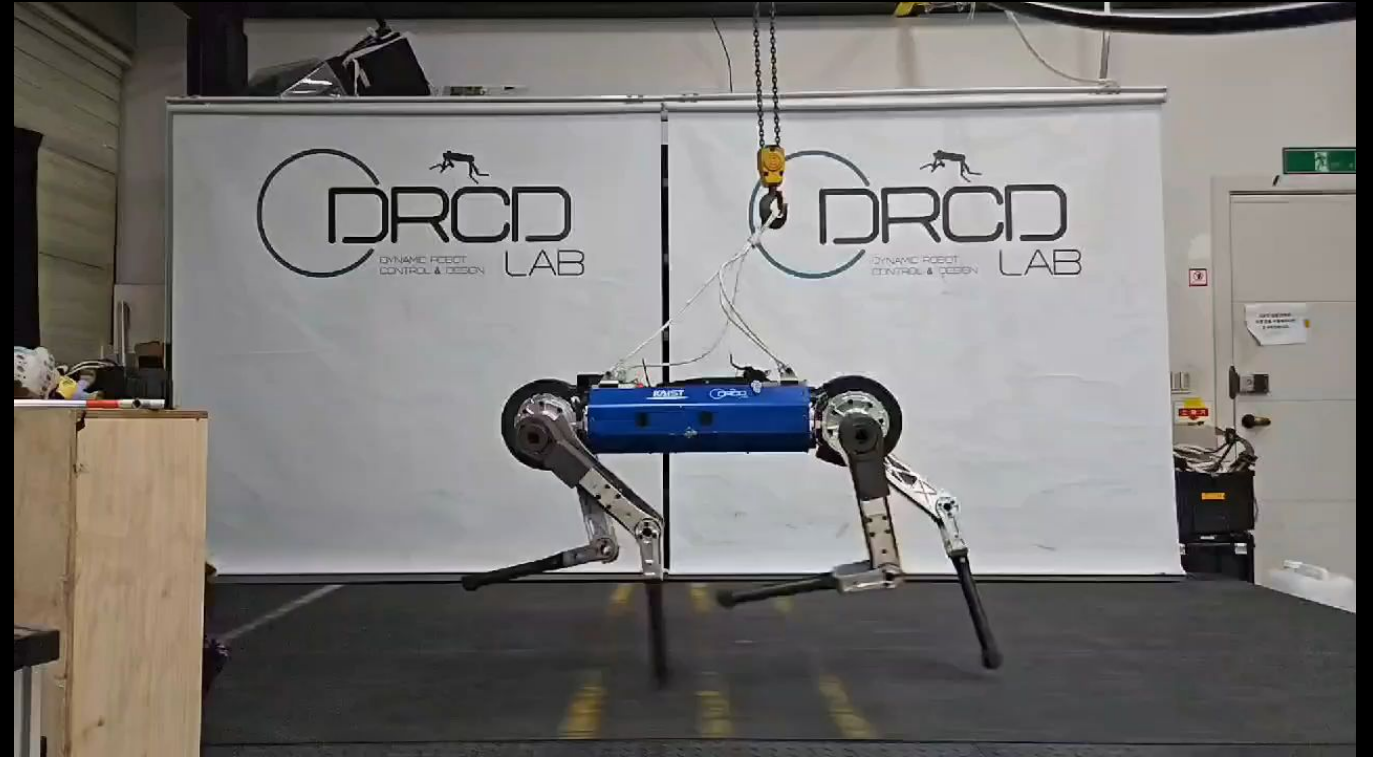
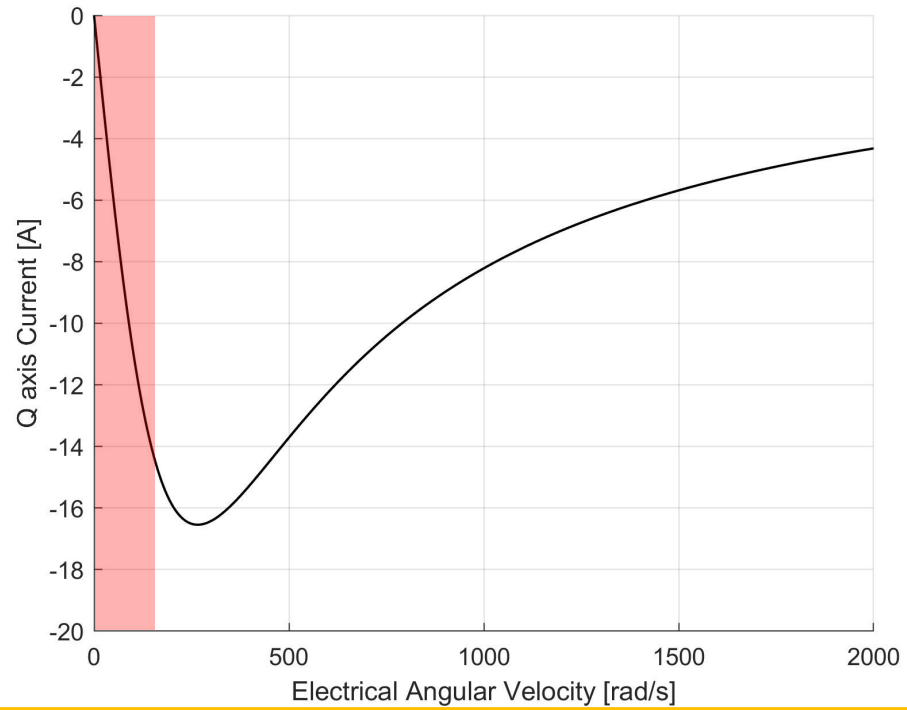
$$\begin{cases} \cancel{V_{ds}^r} = R_s \cancel{i_{ds}^r} + \cancel{L_s \frac{di_{ds}^r}{dt}} - \omega_e L_s i_{qs}^r = 0 \text{ (short)} \\ \cancel{V_{qs}^r} = R_s \cancel{i_{qs}^r} + \cancel{L_s \frac{di_{qs}^r}{dt}} + \omega_e L_s i_{ds}^r + \omega_e \phi_f = 0 \text{ (short)} \end{cases}$$

steady state 일때 0.

$$\begin{aligned} R_s i_{ds}^r - \omega_e L_s i_{qs}^r &= 0 \\ R_s i_{qs}^r + \omega_e L_s i_{ds}^r + \omega_e \phi_f &= 0 \end{aligned} \rightarrow \begin{bmatrix} i_{ds}^r \\ i_{qs}^r \end{bmatrix} = \begin{bmatrix} R_s & -\omega_e L_s \\ \omega_e L_s & R_s \end{bmatrix}^{-1} \begin{bmatrix} 0 \\ -\omega_e \phi_f \end{bmatrix}$$

$$\rightarrow i_{ds}^r(\omega_e) = \frac{-\omega_e^2 \phi_f L_s}{R_s^2 + \omega_e^2 L_s^2}, \quad i_{qs}^r(\omega_e) = \frac{-\omega_e \phi_f R_s}{R_s^2 + \omega_e^2 L_s^2}$$

Problem 3. Resistive Braking



Resistive braking for safety (HOUND2)

Actuator Optimal Design

Lecture 5 : Analytical Calculation of Torque Constant(上)

2026/04/29

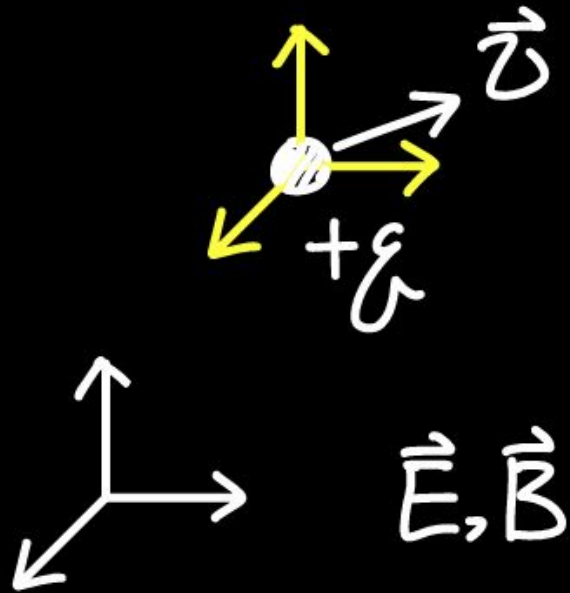
KAIST DRCD lab 송태규 박사과정

Motional EMF revisited

$$: e = Blv$$

Motional EMF revisited : $\mathcal{e} = Blv$

- Lorentz transformation (steady state)

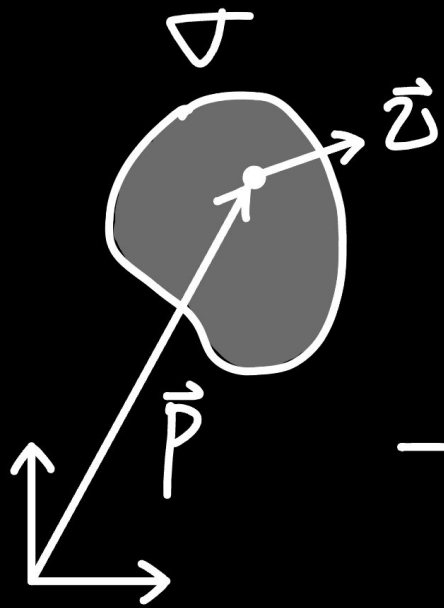


$$\vec{F} = \underbrace{\mathcal{E}(\vec{E} + \vec{v} \times \vec{B})}_{\text{Stationary}} = \underbrace{\mathcal{E}(\vec{E}' + \vec{0} \times \vec{B}')}_{\text{moving}}$$

$$\rightarrow \vec{E}' = \vec{E} + \vec{v} \times \vec{B}$$

Motional EMF revisited : $e = Blv$

- Lorentz transformation (steady state)



$$\vec{J}_b = 0 = \sigma \vec{E}_b \rightarrow \vec{E}_b = 0$$

: steady state condition

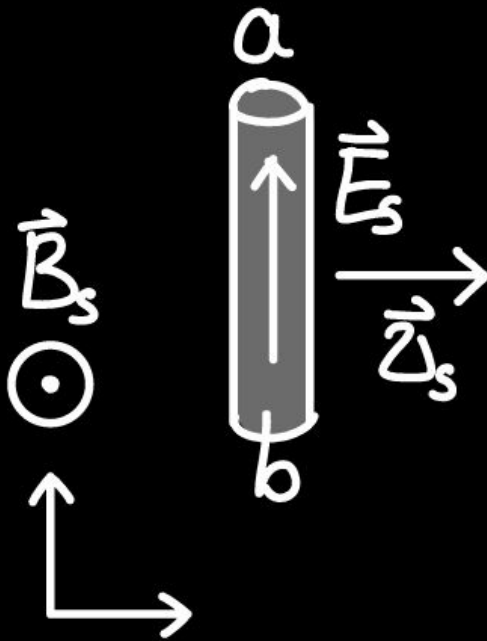
$$\vec{E}_b = 0 = \vec{E}_s + \vec{z}_s \times \vec{B}_s$$

: Lorentz condition

$$\rightarrow \vec{E}_s = \vec{B}_s \times \vec{z}_s$$

Motional EMF revisited : $\mathcal{e} = Blv$

- Lorentz transformation (steady state)



$$\vec{E}_s = -\vec{v}_s \times \vec{B}_s$$

$$\mathcal{e}_{ab} = \int_b^a \vec{E}_s \cdot d\vec{\ell} = \int_b^a \vec{B}_s \times \vec{v}_s \cdot d\vec{\ell} = Blv$$

$$\rightarrow \mathcal{e}_{ab} = Blv$$

[Quiz]

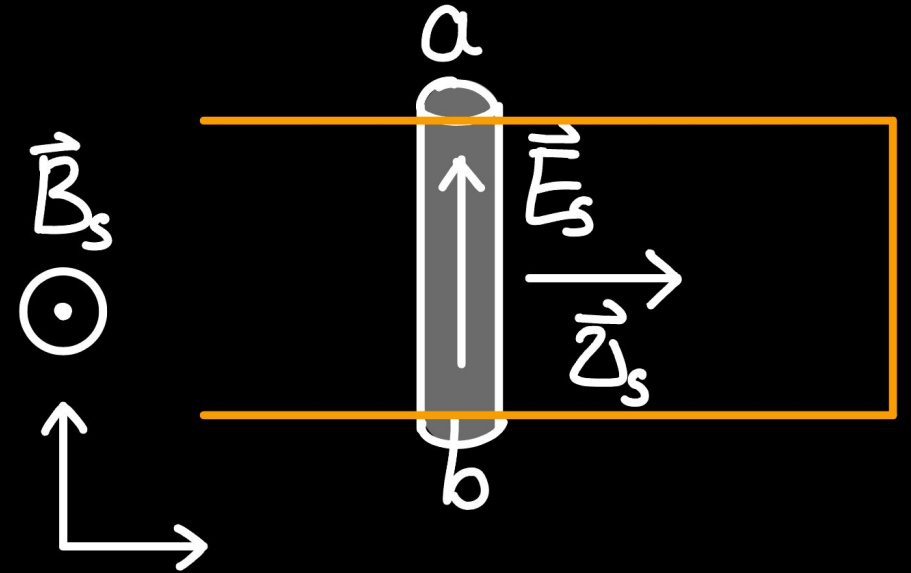
The ends of the rod are open. If the rod has resistance R , does current flow?

Motional EMF revisited : $\mathcal{e} = Blv$

- [Quiz]

What is the difference between definition of EMF and voltage?

(Hint : path)



- [Quiz]

The resistance between the two ends of the rod is R . The rod moves while maintaining contact at both ends with fixed, zero-resistance (orange) rails fixed in the reference frame.

In this case, what are EMF $\mathcal{e}_{ab}$ and voltage v_{ab} ?

Analytical Calculation of Torque Constant

Analytical Calculation of Torque Constant

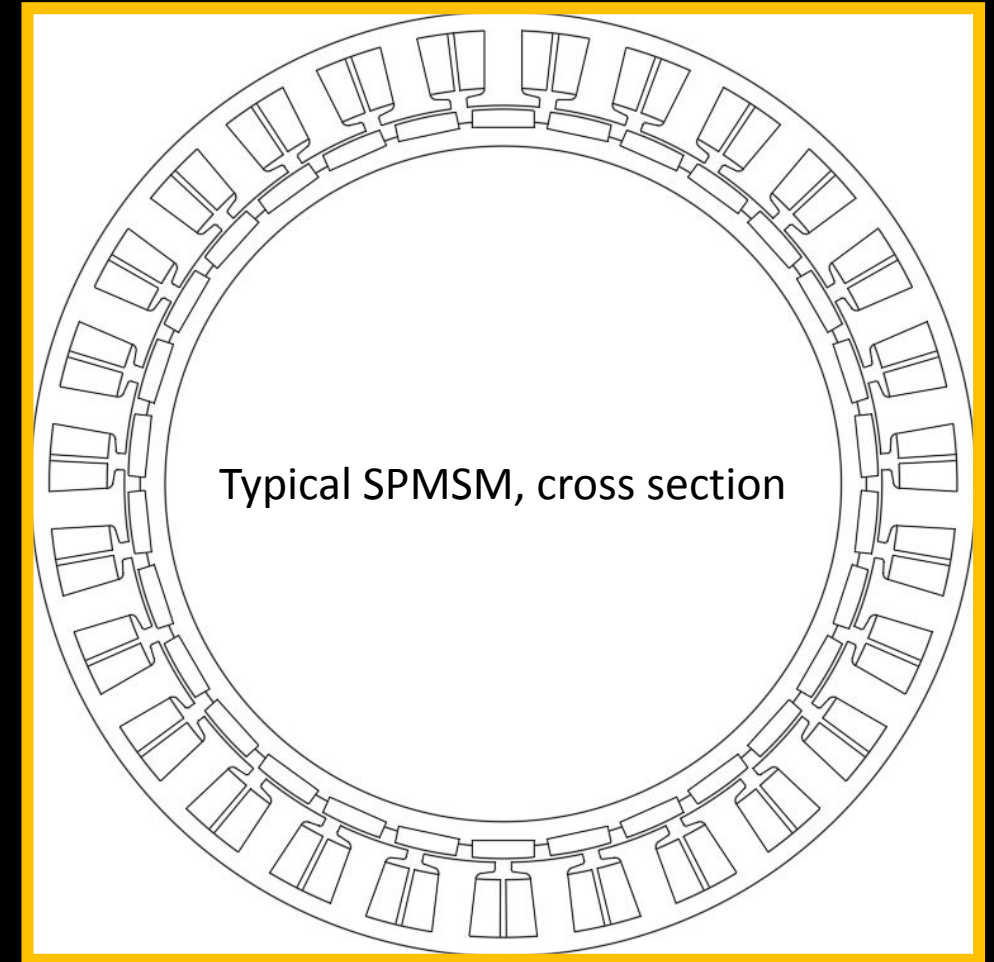
- Purpose

Analytic Calculation of Torque constant with just

1) cross section 2) length 3) magnet grade, **without FEM**.

$$K_t = 3k_w \frac{t}{g} B_r N R l$$

- k_w is winding factor [-]
- t, g is magnet thickness and effective airgap [m]
- B_r is remanence [T] or [N/(A · m)]
- $2N$ is the number of turns per phase [-]
- R is airgap radius [m]
- l is axial length of the motor [m]



Analytical Calculation of Torque Constant

- Main Concept

Torque constant can be derived from the backEMF constant via the '*principle of virtual work*'.

Thus, evaluating backEMF at constant speed gives the torque constant.

- Calculation Procedure

1. Understand '*Magnetic Co-energy*' (W'_m).

2. Verify that $\tau = \frac{\partial W'_m}{\partial \theta}$.

3. Verify that $W'_m = Li^2 + \lambda_f i$.

4. Understand Conductor Density Function.

5. Calculate λ_f by integration of $e = Blv$.

6. Calculate the torque of SPMSM (winding factor = 1).

Analytical Calculation of Torque Constant

- Magnetic Energy (W_m)

The energy stored in a 'magnetic field'.

For simplicity, let's assume a lossless system ($R=0$).

$$\frac{dW_{in}}{dt} = vi = (\cancel{iR} + e)i = ei = \frac{d\lambda}{dt}i$$

$$\Rightarrow dW_{in} = i d\lambda$$

$$dW_{out} = f dx \quad (f: \text{positive when exerted 'by' the system})$$

$$dW_m \triangleq dW_{in} - dW_{out} = i d\lambda - f dx$$

$$\Rightarrow dW_m = i d\lambda - f dx$$

Analytical Calculation of Torque Constant

- Magnetic Co-energy (W'_m)

$$W_m + W'_m \triangleq \lambda i$$

$$dW'_m = \lambda di + i d\lambda - dW_m = \lambda di + f dx$$

$$\Rightarrow dW'_m = \lambda di + f dx$$

$$dW'_m = \frac{\partial W'_m}{\partial i} di + \frac{\partial W'_m}{\partial x} dx = \lambda di + f dx$$

$$\Rightarrow f = \frac{\partial W'_m}{\partial x} \quad \text{with same logic,} \quad \tau = \frac{\partial W'_m}{\partial \theta}$$

Analytical Calculation of Torque Constant

- Magnetic Co-energy (W'_m) with Flux linkage(λ) and Current(i)

$$W'_m = \int_{0,0}^{i,x} dW'_m \triangleq \int_{0,0}^{i,x} \lambda di + f dx = \int_{0,0}^{0,x} \lambda \cancel{di} + f dx + \int_{0,x}^{i,x} \lambda di + f \cancel{dx}$$

Cogging Torque

Ignore Cogging Torque

$$= \int_{0,0}^{0,x} f dx + \int_{0,x}^{i,x} \lambda di \approx \int_0^i \lambda(x) di = \int_0^i L(x)i + \lambda_f(x) di = \frac{1}{2}L(x)i^2 + \lambda_f(x)i$$

$$\Rightarrow W'_m = \lambda_f(x)i$$

$$\lambda = Li + \lambda_f$$

by rotor magnet

by current flowing in stator winding

Analytical Calculation of Torque Constant

- Magnetic Co-energy (W'_m) with Flux linkage(λ) and Current(i)

$$W'_m = \frac{1}{2}L(x)i^2 + \lambda_f(x)i$$

$$\tau = \frac{\partial W'_m}{\partial \theta} = \underbrace{\frac{\partial L}{\partial \theta} i^2}_{\text{Reluctance Torque}} + \underbrace{\frac{\partial \lambda_f}{\partial \theta} i}_{\text{Field-alignment Torque}}$$

$$\frac{\partial L}{\partial \theta} \approx 0 \text{ for SPMSM (NOT for IPMSM)}$$

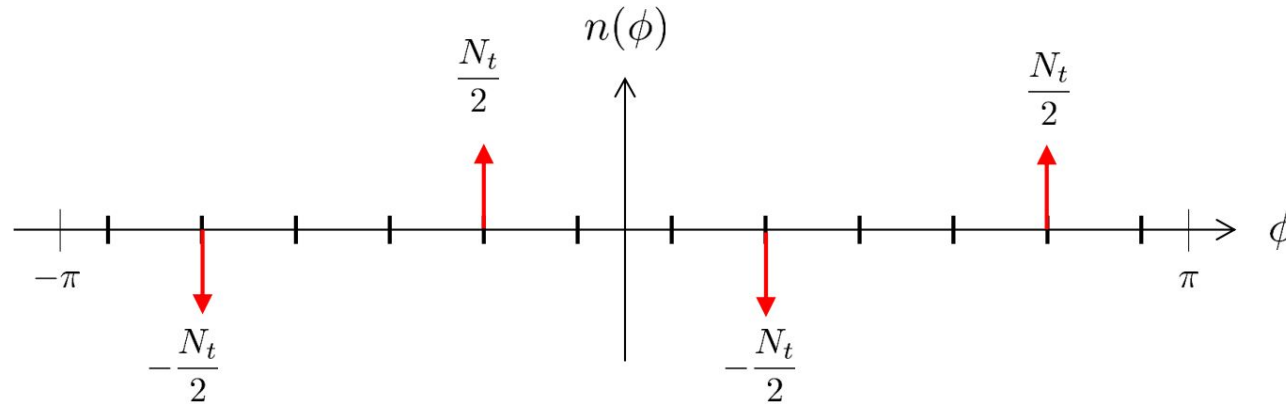
$$\Rightarrow \tau = \frac{\partial \lambda_f}{\partial \theta} i \text{ (SPMSM)}$$

Analytical Calculation of Torque Constant

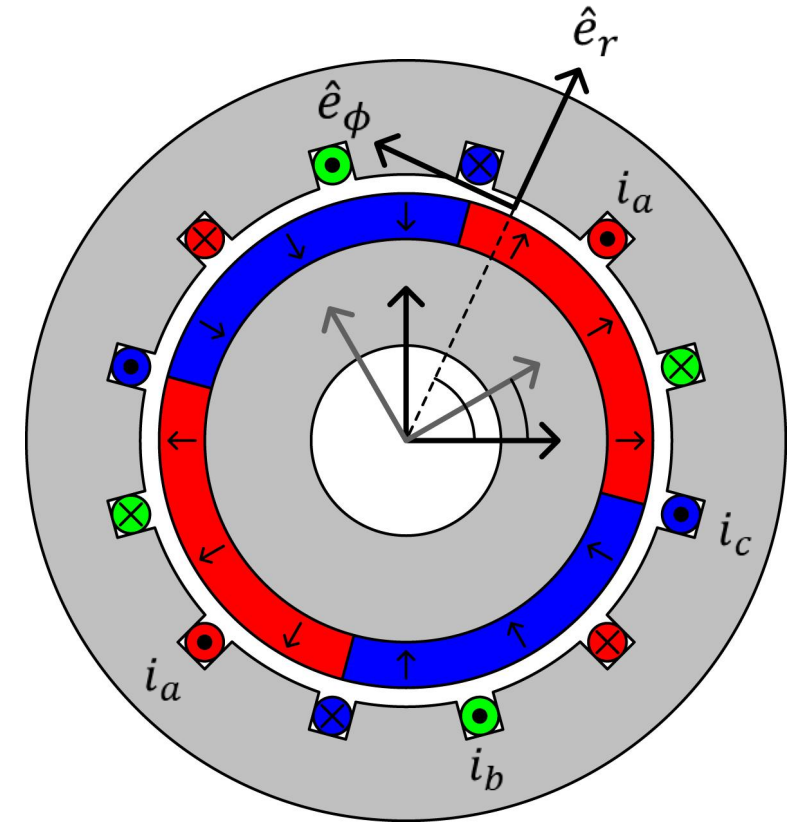
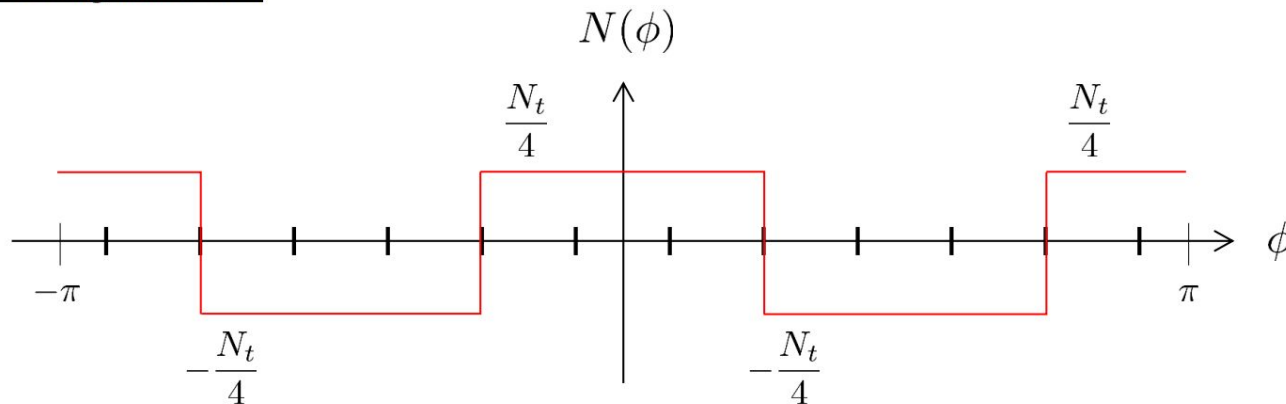
- Conductor Density Function ($n(\phi)$) & Winding Function ($N(\phi)$)

Number of conductors within $d\phi$ at ϕ is $n(\phi)d\phi$.

Conductor density function



Winding function



$2N_t$ is the number of turns per phase [-]

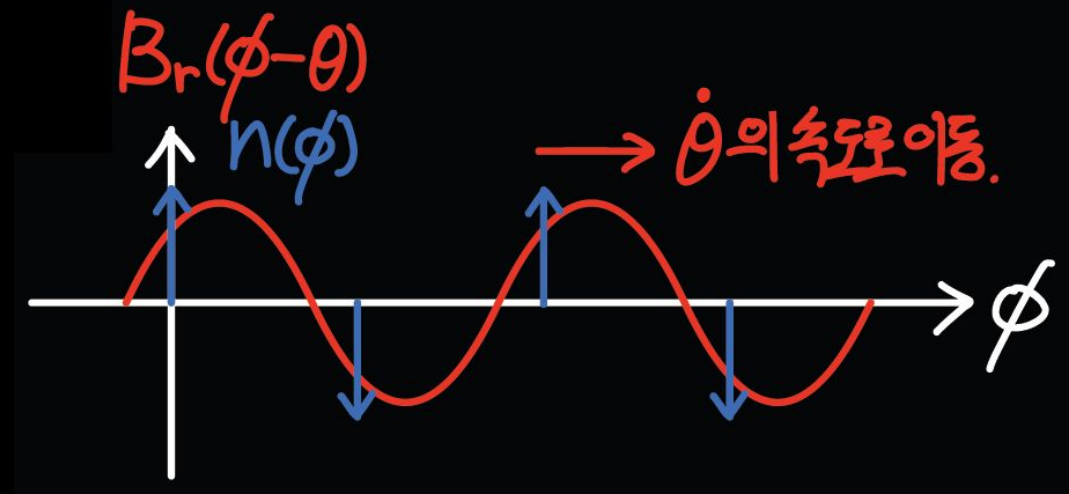
Analytical Calculation of Torque Constant

- EMF (e) from $e = Blv$

A상 양단의 emf e :

$$e = \sum_i (n(\phi_i) \Delta\phi_i) B_r(\phi_i - \theta) l (R\dot{\theta})$$

$$= Rl\dot{\theta} \int_0^{2\pi} B_r(\phi - \theta) n(\phi) d\phi = Rl\dot{\theta} \int_0^{2\pi} B_r(\phi) n(\phi + \theta) d\phi$$



B_r : Magnetic Field Density, radial

Analytical Calculation of Torque Constant

- Flux Linkage (λ_f)

$$\lambda_f = C \int_0^{2\pi} B_r(\phi - \theta) N(\phi) d\phi = C \int_0^{2\pi} B_r(\phi) N(\phi + \theta) d\phi \text{ 풀이라고 하자.}$$

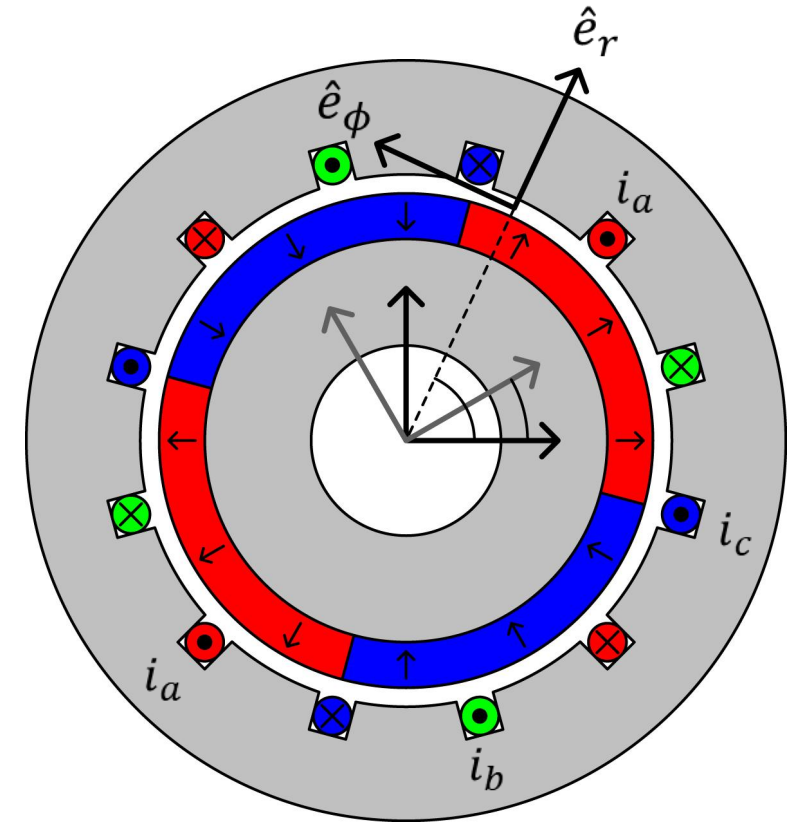
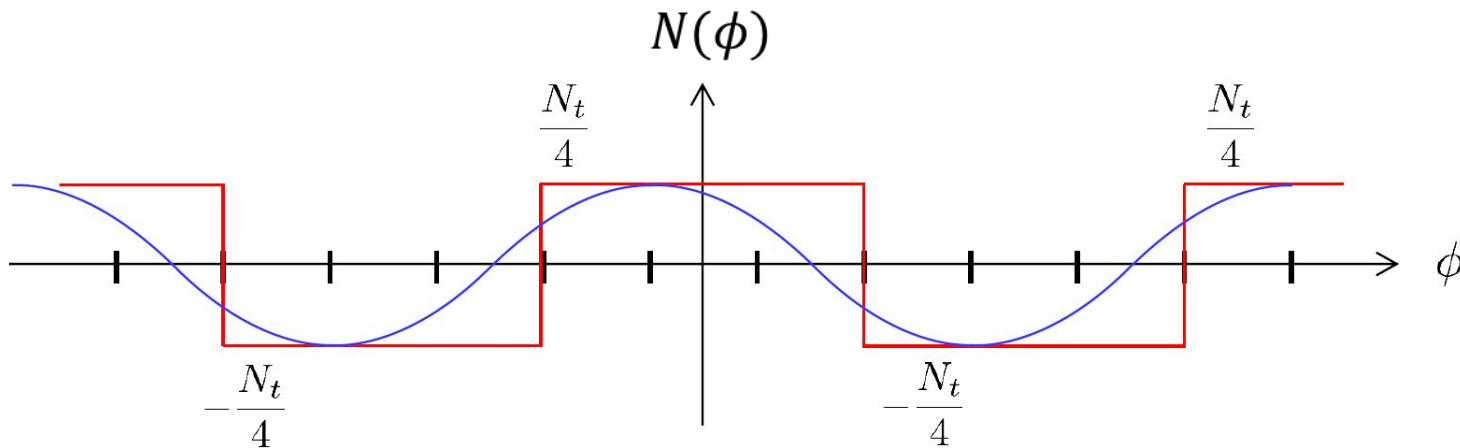
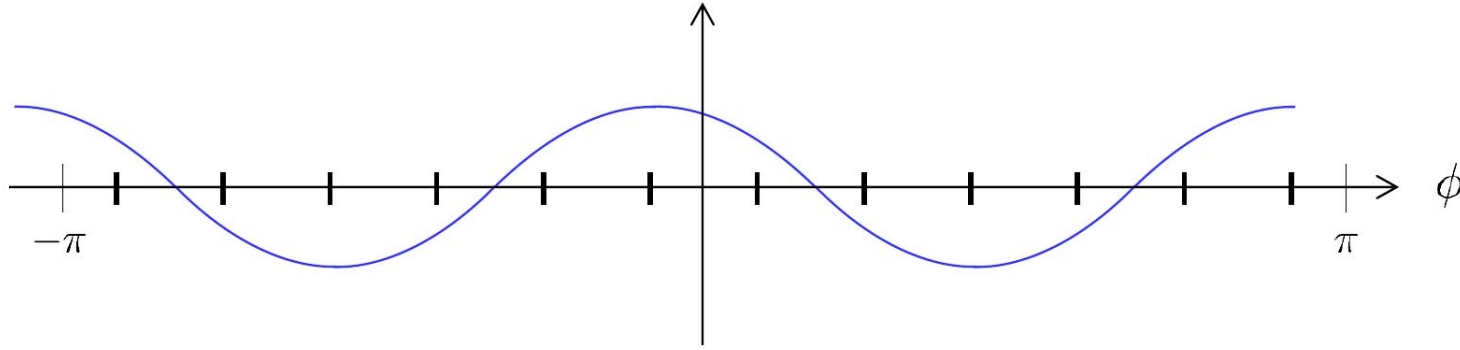
$$e = \frac{d\lambda_f}{dt} = \frac{d\lambda}{d\theta} \frac{d\theta}{dt} = C \dot{\theta} \int_0^{2\pi} B_r(\phi) \frac{d}{d\theta} N(\phi + \theta) d\phi = C \dot{\theta} \int_0^{2\pi} B_r(\phi) n(\phi + \theta) d\phi$$

$$\text{따라서 } \lambda_f(\theta) = R_l \int_0^{2\pi} B_r(\phi) N(\phi + \theta) d\phi = R_l \int_0^{2\pi} B_r(\phi - \theta) N(\phi) d\phi$$

Analytical Calculation of Torque Constant

- $B_r(\phi - \theta)$ & $N(\phi)$

$$B_r(\phi - \theta) = B_m \cos(2(\phi - \theta))$$



$2N_t$ is the number of turns per phase [-]

Analytical Calculation of Torque Constant

- Sinusoidal Magnetization Assumption

편의상 $B_r(\phi) = B_m \cos p\phi$. 삼각파 자화를 가정하자. 실제로는 trapezoidal 함.

$$W_m'(\theta) = \lambda_f(\theta) i$$

$$= i R \ell \int_0^{2\pi} B_r(\phi - \theta) N(\phi) d\phi$$

$$= i R \ell B_m \int_0^{2\pi} \cos(p(\phi - \theta)) N(\phi) d\phi$$

Analytical Calculation of Torque Constant

- Torque produced by Phase A

$$\begin{aligned} W'_m(\theta) &= i R l B_m \int_0^{2\pi} \cos(p(\phi - \theta)) \frac{N}{2p} \text{square}(p\phi) d\phi \\ &= 2i R l B_m \frac{N}{p} \cos p\theta \end{aligned}$$

$$\begin{aligned} \tau_a &= \frac{\partial W'_m}{\partial \theta} \\ &= -2i_a R l B_m N \sin p\theta \end{aligned}$$

Analytical Calculation of Torque Constant

- Total Torque

$$\tau_k = -2i_k R l B_m N \sin p(\theta - \theta_k) \quad (\theta_a = 0, \theta_b = -\frac{2\pi}{3}, \theta_c = +\frac{2\pi}{3})$$

$$\tau = \sum_k \tau_k = 2Rl B_m N i_g \sum_k \sin^2 p(\theta - \theta_k) = 3B_m Rl N i_g$$

$$i_k = -i_g \sin p(\theta - \theta_k)$$

$$\tau = 3\left(\frac{t}{g} B_r\right) N R l i_g$$

$$\Rightarrow K_t = 3k_w \frac{t}{g} B_r N R l$$

Analytical Calculation of Torque Constant

- Phase Current

$$\cdot \times \cdot \sin^2 \alpha + \sin^2(\alpha - \frac{2\pi}{3}) + \sin^2(\alpha + \frac{2\pi}{3}) = \frac{3}{2}$$

$$\circ i_k = -i_g \sin(\theta - \theta_k)$$

$\cdot \times \cdot$ Park-Clark transformation에 의해,

$$\begin{bmatrix} i_d \\ i_g \end{bmatrix} \triangleq \frac{2}{3} \begin{bmatrix} \cos \theta & \cos(\theta - \frac{2\pi}{3}) & \cos(\theta - \frac{4\pi}{3}) \\ -\sin \theta & -\sin(\theta - \frac{2\pi}{3}) & -\sin(\theta - \frac{4\pi}{3}) \end{bmatrix} \begin{bmatrix} i_a \\ i_b \\ i_c \end{bmatrix}$$

$i_d = 0$ 일 때 $i_k = I_o \sin(\theta - \theta_k)$ 이면

$$i_g = -\frac{2}{3} I_o \left\{ \sin^2 \theta + \sin^2(\theta - \frac{2\pi}{3}) + \sin^2(\theta + \frac{2\pi}{3}) \right\}$$

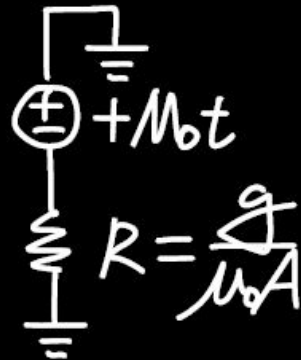
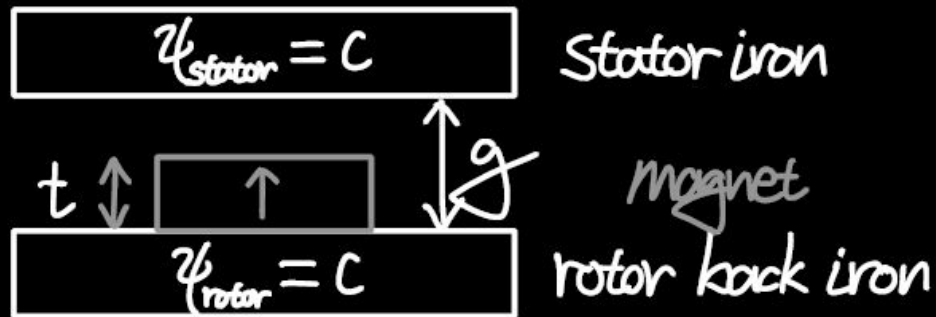
$$i_g = -I_o. \text{ 따라서 } i_d = 0, i_g \text{가 일정할 때 } i_k = -i_g \sin(\theta - \theta_k)$$

Analytical Calculation of Torque Constant

- Amplitude of $B_r(\phi)$

$$\circ B_m = \frac{t}{g} B_r$$

왜 $\psi_{\text{stator}} = \psi_{\text{rotor}}$ 인지는 Proof A, B 참고



$$\phi = \frac{M_0 t}{\frac{g}{\mu_0 A}} = \frac{t}{g} \mu_0 M_0 A = BA \rightarrow B = \frac{t}{g} \mu_0 M_0 = \frac{t}{g} B_r$$

B_r : 제로에서 제공.